

col 703/
7203

Natural Deduction

30 July

$$\frac{\phi_1 \quad \phi_2}{\phi_1 \wedge \phi_2} \quad \wedge_i$$

$$\frac{\phi_1 \wedge \phi_2}{\phi_1} \quad \wedge_e \quad \frac{\phi_1 \wedge \phi_2}{\phi_2} \quad \wedge_e$$

$$\boxed{\begin{array}{l} \phi \text{ assumed} \\ \vdots \\ \psi \end{array}} \quad \Rightarrow_i \quad \frac{}{\phi \Rightarrow \psi}$$

$$\frac{\phi}{\neg \neg \phi}$$

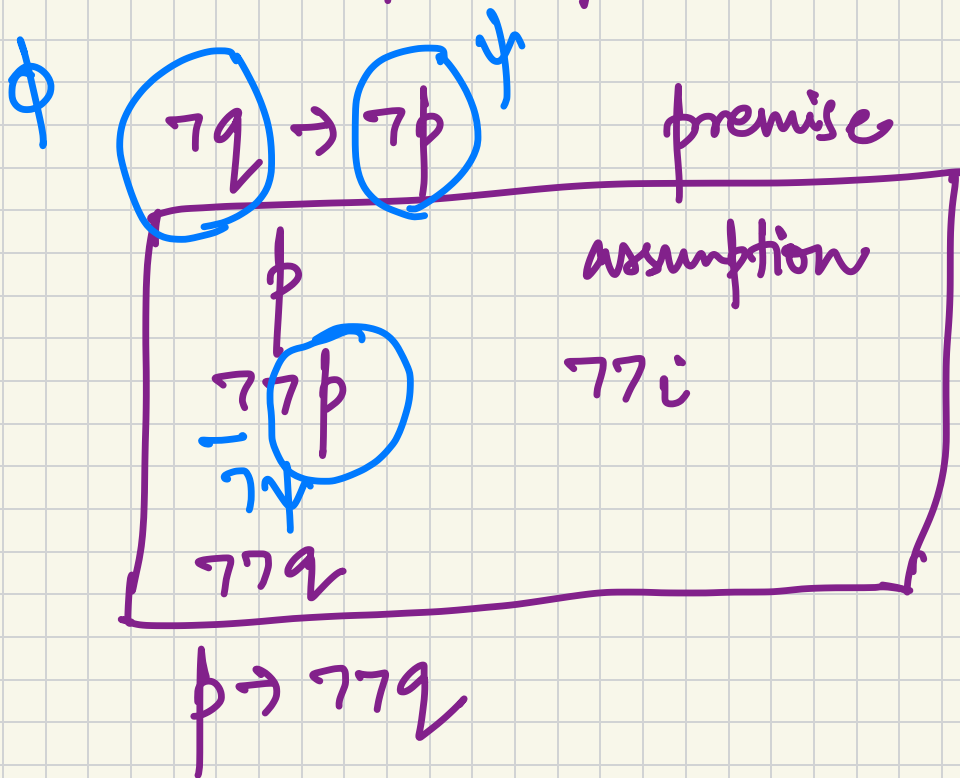
$$\frac{\neg \neg \phi}{\phi}$$

$$\frac{\phi \quad \phi \Rightarrow \psi}{\psi} \quad \text{MP}$$

$$\frac{\phi \Rightarrow \psi \quad \neg \psi}{\neg \phi} \quad \text{MT}$$

$$\neg q \rightarrow \neg p$$

$$\vdash p \rightarrow \neg\neg q$$



$$\vdash (p \rightarrow p)$$

p	assum
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$$p \rightarrow p$$

$$p \wedge q \rightarrow r \vdash p \rightarrow (q \rightarrow r)$$

0.	$p \wedge q \rightarrow r$	premise
1.	p	assumption
2.	q	assumption
3.	$p \wedge q$	$\wedge i$ 1, 2
4.	r	MP 3, 0
	$(q \rightarrow r)$	
	$p \rightarrow (q \rightarrow r)$	

$$p \rightarrow (q \rightarrow r) \vdash p \wedge q \rightarrow r$$

0.	$p \rightarrow (q \rightarrow r)$	
1.	$p \wedge q$	assumptions
2.	p	$\wedge e_1, 1$
3.	$q \rightarrow r$	MP 2, 0
4.	q	$\wedge e_2, 1$
5.	r	MP, 4, 3
$p \wedge q \rightarrow r$		

$p \rightarrow q \quad \vdash \quad p \wedge r \rightarrow q \wedge r$

$p \wedge r$

p

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assumes

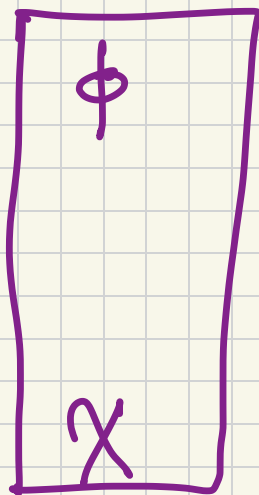
$\wedge e$

Disjunction

$$\frac{\phi}{\phi \vee \psi} \quad v_{i_1}$$

$$\frac{\psi}{\phi \vee \psi} \quad v_{i_2}$$

$\phi \vee \psi$



v_e

χ

$p \vee q \vdash q \vee p$

$p \vee q$

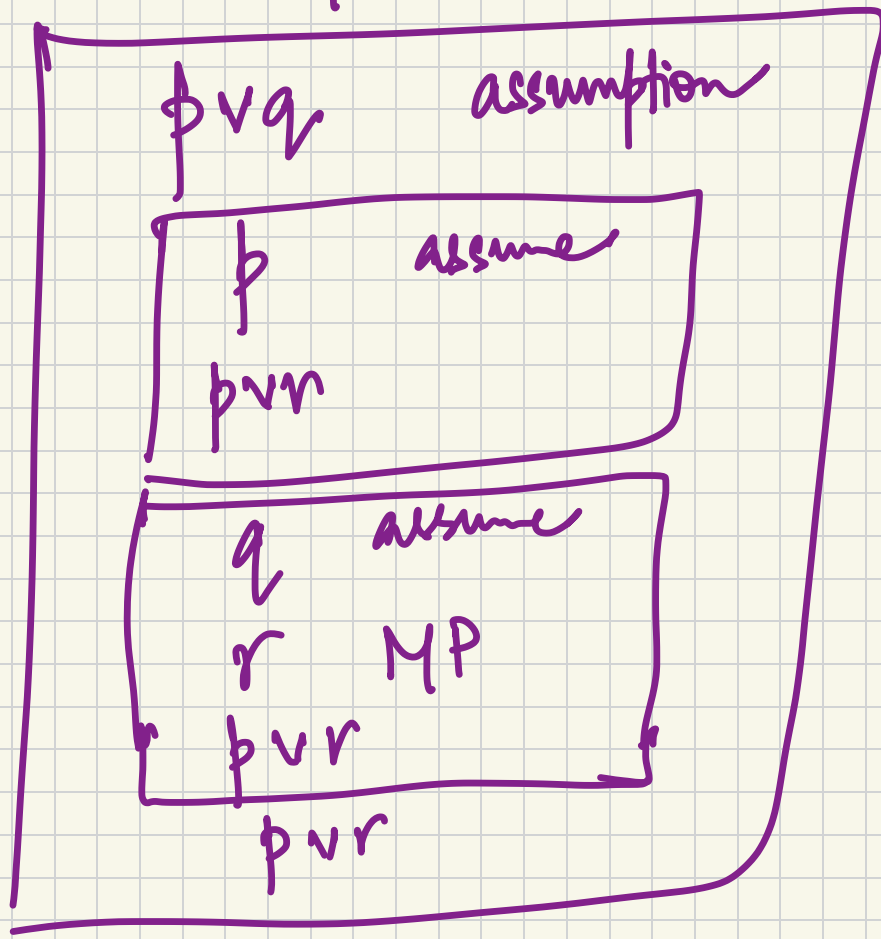
p
 $q \vee p$

q
 $q \vee p$

$q \vee p$

$p \vee q$

$$q \rightarrow r \quad \vdash \quad p \vee q \rightarrow p \vee r$$

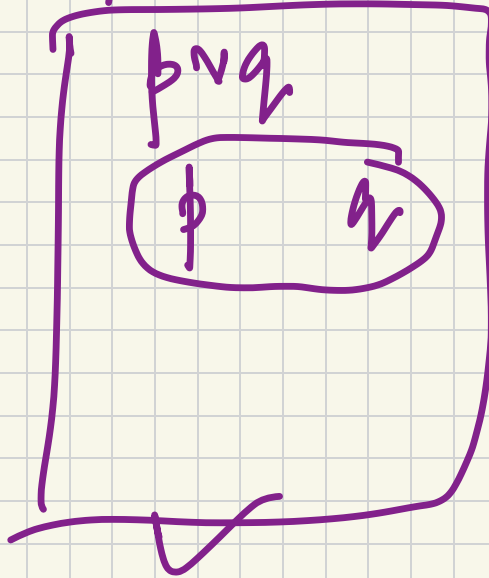


$$p \vee q \rightarrow p \vee r$$

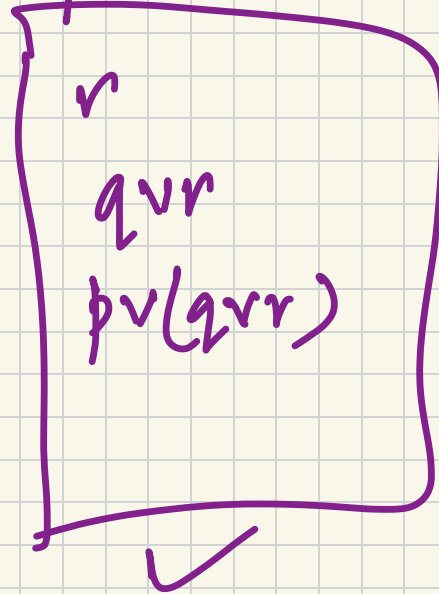
$(p \vee q) \vee r \vdash$

$p \vee (q \vee r)$

$(p \vee q) \vee r$



premise

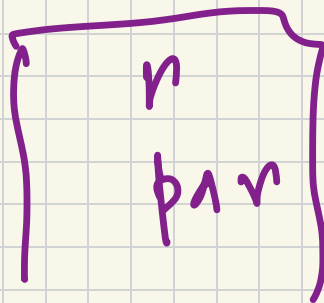
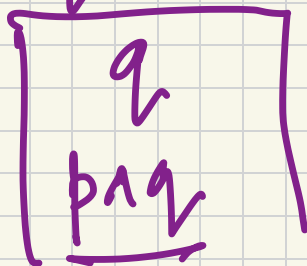


$$p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$$

$p \wedge (q \vee r)$ premise

- p

- $q \vee r$



Exercise

$$p \vee (q \wedge r) \vdash (p \vee q) \wedge (p \vee r)$$

What about negations?

Contradictions

$$\left. \begin{array}{l} \phi \wedge \neg \phi \\ \neg \phi \wedge \phi \end{array} \right\}$$

denoted as $\perp$ (bottom)

Contradictions allow us to derive anything.

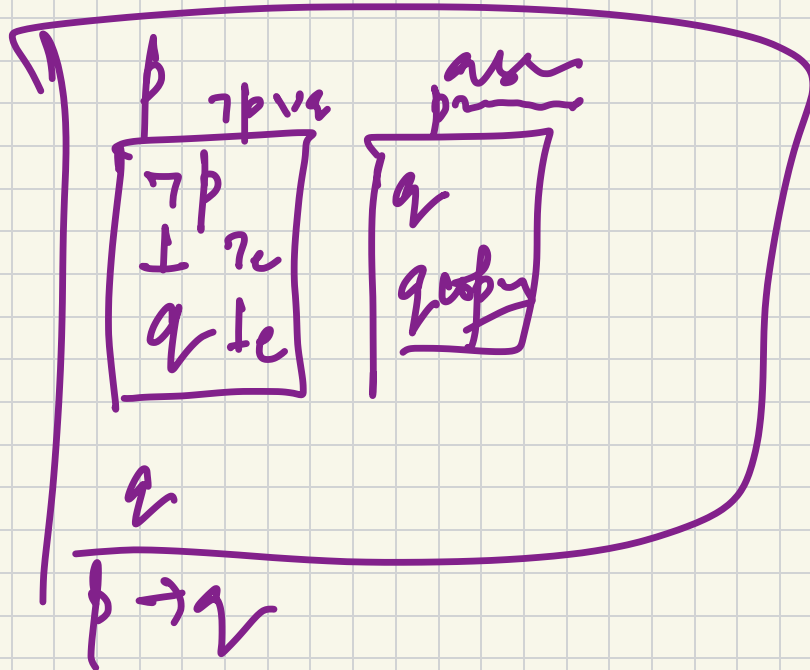
$$\frac{\perp}{\phi} \quad \perp e$$

$$\frac{\phi \quad \neg\phi}{\perp} \quad \neg e$$

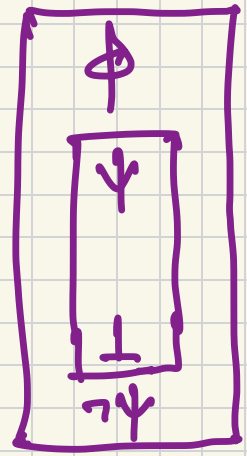
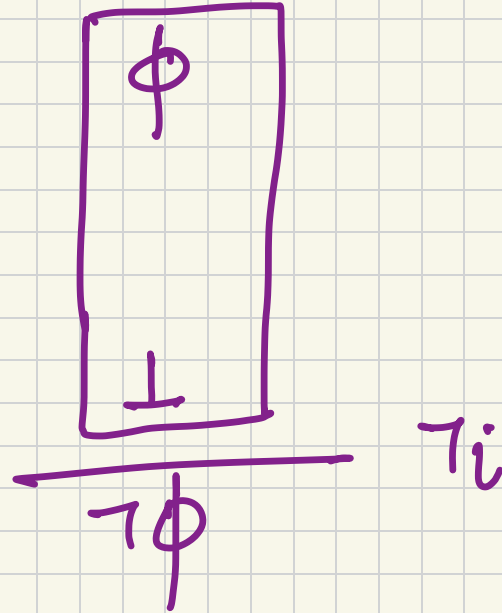
$$p \wedge \neg p \vdash q$$

Exercise

$$\neg p \vee q \vdash p \rightarrow q$$



Introduce negations



Exercises i) $p \rightarrow q, p \rightarrow \neg q \vdash \neg p$

ii) $p \rightarrow \neg p \vdash \neg p$

iii) $p \wedge \neg q \rightarrow r, \neg r, p \vdash q$